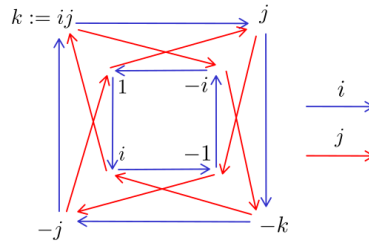




1. Recall the Cayley graph of the quaternion group Q_2 with generating set $S = \{i, j\}$.



From this graph, it turns out that $Q_2 = \{\pm 1, \pm i, \pm j, \pm k\}$ and $i^2 = j^2 = k^2 = ijk = -1$. Write down the composition table for Q_2 .

2. Let $\mathbb{H}$ denote the four dimensional vector space $\mathbb{R} + \mathbb{R}i + \mathbb{R}j + \mathbb{R}k$. Show that the composition law of Q_2 , by imposing associativity and distributivity, extends to the composition rule $*$ in on $\mathbb{H}$ as follows:

For $a + bi + cj + dk, x + yi + zj + wk \in \mathbb{H}$,

$$\begin{aligned} (a + bi + cj + dk) * (x + yi + zj + wk) = & (ax - by - cz - dw) \\ & + (ay + bx + cw - dz)i \\ & + (az - bw + cx + dy)j \\ & + (aw + bz - cy + dx)k \end{aligned}$$

The $*$ above is called the *quaternionic multiplication*, and the pair $(\mathbb{H}, *)$ is called the *real quaternion algebra*.

3. Is $\mathbb{H}$ a group under $*$? What about the set of nonzero elements of $\mathbb{H}$?
4. Show that the set of unit quaternions $\{q \in \mathbb{H} : \|q\| = 1\}$ forms a group under quaternionic multiplication. This group is also called *3-sphere group*.
5. Let $q = a + bi + cj + dk$ be a unit quaternion. Let

$$A_q = \begin{pmatrix} a + ib & -c + id \\ c + id & a - ib \end{pmatrix}$$

Show that A is a unitary matrix, that is $A_q^\dagger A_q = I$, where $A_q^\dagger$ denotes the conjugate transpose of A_q . Further, show that $\det(A_q) = 1$.